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# R V COLLEGE OF ENGINEERING®

**Course: EXPERIENTIAL LEARNING**
**Course Code: IS266TEO**

# "StarTrack AR – Augmented Reality Based International Space Station Tracking and Astronomy Assistant"



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**2025–2026**

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# CERTIFICATE

Certified that the project work titled **StarTrack AR – Augmented Reality Based International Space Station Tracking and Astronomy Assistant** is carried out by **B Vinayaka Aili (1RV23AI023), Rahul N M (1RV23CS190)** who are bonafide students of **RV College of Engineering**, Bengaluru, in fulfilment of the Experiential Learning Project during the academic year **2025–2026**.

It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of experiential learning work prescribed by the institution for the said degree.

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# DECLARATION

We, **B Vinayaka Aili, Rahul N M**, students of sixth semester B.E., **RV College of Engineering**, Bengaluru, hereby declare that the Experiential Learning project titled **StarTrack AR – Augmented Reality Based International Space Station Tracking and Astronomy Assistant** has been carried out by us and submitted in fulfilment of the Experiential Learning Project during the academic year **2025–2026**.

We also declare that any **Intellectual Property Rights** generated out of this project carried out at RVCE will be the property of **RV College of Engineering, Bengaluru** and we will be one of the authors of the same.

Place: Bengaluru Date:

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**2. RAHUL N M (1RV23CS190)**

# ABSTRACT

The International Space Station (ISS) is the largest human-made object in low Earth orbit and remains one of the most visible and inspiring symbols of contemporary space exploration. Despite its visibility to the unaided eye under suitable conditions, locating the ISS and other celestial objects in real time remains a challenging task for amateur sky-watchers and students, who typically lack convenient access to the orbital data, directional guidance, and contextual astronomical knowledge required to do so. StarTrack AR – Augmented Reality Based International Space Station Tracking and Astronomy Assistant is an Android application developed to address this gap by combining live satellite tracking, Augmented Reality (AR) visualisation, multi-sensor fusion, and an AI-powered conversational assistant within a single, beginner-friendly mobile experience.

The application is developed using Android Studio with Java and XML, and integrates the device GPS, accelerometer, and magnetometer sensors with the smartphone camera to provide real-time directional guidance toward the ISS. Live orbital position data for the ISS is retrieved from the N2YO satellite tracking API, while Sun, Moon, and planetary position data is obtained from AstronomyAPI. Sky visibility and cloud-cover information is sourced from the OpenWeatherMap API, enabling the application to inform users of optimal viewing conditions. An AI astronomy chatbot, powered by the Google Gemini API, allows users to ask natural-language questions about space and receive beginner-friendly explanations. A dedicated notification module alerts users of upcoming ISS passes that are visible from their location.

The core technical contribution of the system lies in its coordinate transformation pipeline, which converts the geographic position of the ISS and the user into azimuth and elevation angles, and continuously compares these against the live orientation of the device (computed through sensor fusion of the accelerometer and magnetometer) to render an Augmented Reality overlay that guides the user to point their camera directly at the satellite. The application is organised into six principal modules: Home, AR Camera, ISS Tracking, Tonight's Sky, Chatbot, and Notification, each responsible for a distinct functional concern within the overall architecture.

This report presents the problem context, review of existing astronomy and satellite-tracking applications, design methodology, implementation details, and evaluation results of StarTrack AR.

# TABLE OF CONTENTS

# Chapter 1
# PROBLEM STATEMENT

The International Space Station orbits the Earth at an altitude of approximately 408 kilometres, completing one full revolution roughly every 92 minutes and travelling at a speed of nearly 27,600 kilometres per hour. Under suitable lighting and weather conditions, the ISS is visible to the naked eye from the ground as a fast-moving, bright point of light, making it one of the most accessible objects in observational astronomy for the general public. Despite this accessibility, very few people are able to reliably locate the ISS, or other celestial objects such as planets, in the night sky without specialised knowledge of orbital mechanics, compass direction, and elevation angles.

Existing astronomy and satellite-tracking applications address only fragments of this problem. Star-mapping applications such as Sky Map provide a generic augmented overlay of fixed stars but do not track artificial satellites such as the ISS. Dedicated satellite-tracking services such as the NASA Spot the Station tool and N2YO.com provide accurate pass-time predictions but operate primarily as static, map-based or list-based web interfaces, without any camera-based augmented reality guidance to help a user physically orient their device toward the satellite. Premium planetarium applications such as Stellarium Mobile provide rich celestial simulation but place AR-based live-sky features behind a paywall and present an interface that can be intimidating for first-time users. None of the surveyed applications combine real-time ISS tracking, AR camera overlay, multi-sensor directional guidance, and an interactive AI assistant within a single, beginner-friendly mobile application.

This fragmentation creates a tangible barrier to public engagement with space science. A student or hobbyist who wishes to observe the ISS must typically consult one application or website for pass-time predictions, manually estimate compass direction and elevation using a separate compass application, and have no readily available means of asking follow-up questions about what they are observing. The absence of an integrated, sensor-fused, AR-guided, AI-assisted solution represents a clear and addressable gap in the current landscape of mobile astronomy tools.

The problem addressed by this project may therefore be stated as follows: there is no unified, beginner-friendly Android application that provides real-time Augmented Reality based directional guidance to the International Space Station and other celestial objects, fusing GPS, accelerometer, and magnetometer data with live satellite and astronomical APIs, while also offering an AI-powered conversational assistant for contextual astronomy education. StarTrack AR is proposed and developed in this project to resolve this gap by delivering a single Android application that performs real-time ISS tracking, renders an AR overlay on the live camera feed, computes precise directional guidance through multi-sensor fusion, aggregates astronomical and weather data into a unified “Tonight's Sky” dashboard, and integrates a generative-AI chatbot for interactive learning, thereby lowering the barrier to entry for amateur space observation and astronomy education.

# Chapter 2
# EXISTING SYSTEM

A range of mobile and web-based astronomy and satellite-tracking tools are currently available to the public, each addressing a subset of the functionality proposed for StarTrack AR. This chapter reviews the most prominent existing systems, summarises relevant academic literature on Augmented Reality astronomy applications and sensor-based orientation tracking, and identifies the specific gaps that motivate this project.

## 2.1 Survey of Existing Applications

Google Sky Map and similar star-mapping applications use the device camera together with accelerometer and magnetometer readings to overlay the names of stars and constellations on the live camera feed. While effective for general stargazing, these applications display only fixed celestial bodies and have no awareness of artificial satellites such as the ISS, nor do they provide any AI-based assistance.

The official NASA mobile application and the Spot the Station service provide accurate ISS pass-time notifications based on orbital propagation from NORAD Two-Line Element (TLE) data. However, neither offers an AR camera mode; users are informed only of the date, time, and approximate compass direction of an upcoming pass, and must independently translate this information into a physical viewing direction.

Stellarium Mobile is a feature-rich planetarium application offering an accurate, GPU-rendered simulation of the night sky, including planets, deep-sky objects, and satellites. Its AR-based live-sky mode, however, is restricted to a paid subscription tier, and the interface presents a steep learning curve for first-time or younger users, owing to the density of astronomical information and configuration options presented simultaneously.

Heavens-Above is a long-established web-based service that computes highly accurate visible-pass predictions for the ISS and other satellites using orbital mechanics, but it is accessible only through a browser and offers no native mobile or AR experience. SkyView Lite and other lightweight compass-based stargazing applications provide sensor-fused sky navigation for stars and planets but do not track the ISS or any other artificial satellite, and offer no conversational or educational assistant.

## 2.2 Literature Review

In addition to the commercial applications surveyed above, the design of StarTrack AR draws on published research in four related areas: Augmented Reality in astronomy education, mobile satellite-tracking systems, sensor-based orientation tracking for AR, and AI-assisted educational interfaces. Table 2.1 summarises the key literature consulted during the design of this project.

| Sl. | Paper / Source | Journal / Conference (Year) | Key Points | Limitations | Relevance to StarTrack AR |
|---|---|---|---|---|---|
| 1 | A survey of augmented reality (Azuma, R.) | Presence: Teleoperators and Virtual Environments (1997) | Establishes the foundational definition and taxonomy of AR systems, including registration and real-world overlay principles | Predates modern smartphone sensor hardware; no mobile or astronomy-specific context | Provides the conceptual basis for overlaying ISS and celestial markers on the live camera feed used in the AR Camera Module |
| 2 | Scoping the Window to the Universe: Design Considerations and Expert Evaluation of an AR Mobile Application for Astronomy Education | Springer, EC-TEL / related proceedings (2017) | Presents design and pupil/teacher evaluation of a mobile AR astronomy application for primary education | Evaluated only with a small teacher panel; lacks live satellite or AI-assistant integration | Validates AR as a pedagogically effective medium for astronomy, supporting the Educational Mode design rationale in StarTrack AR |
| 3 | Augmented Reality Application for Solar System Learning: A Research in Progress | IEEE Conference Publication (2018) | Proposes a mobile AR application for visualising the solar system to aid classroom teaching | Limited to static solar-system models; no live orbital data or directional guidance | Confirms feasibility of AR-based celestial visualisation, extended in this project to live, real-time ISS tracking |
| 4 | A Gamified Mobile Augmented Reality System for the Teaching of Astronomical Concepts (PlanetarySystemGO) | IEEE Conference Publication (2019) | Describes a gamified AR system with a web-based instructor back end for teaching astronomy concepts | Focused on gamification for classroom use; no satellite tracking or sensor-fusion-based pointing guidance | Supports the inclusion of an engaging, beginner-friendly interface, while StarTrack AR additionally targets independent, real-world sky observation |
| 5 | Exploring the Stars: How Augmented Reality Transforms the Teaching of Astronomic Concepts in School | IEEE Conference Publication (2024) | Builds a Unity-based AR system model for visualising planetary and stellar motion in a school setting | Simulation-based rather than live-data-driven; no integration with real APIs or live satellite feeds | Reinforces the use of AR as an engagement tool, while StarTrack AR extends this with live N2YO and AstronomyAPI data |
| 6 | A sensor fusion method for smart phone orientation estimation | Embedded Systems Design literature (2013) | Proposes a drift-and-noise-removal filter combining gyroscope, magnetometer, and accelerometer data for robust orientation estimation | Requires gyroscope data, which is not used in the present accelerometer-and-magnetometer-only implementation; | Directly informs the sensor fusion approach used to compute device azimuth and pitch in the StarTrack AR Sensor Fusion Module, and is referenced in the |

| Sl. | Paper / Source | Journal / Conference (Year) | Key Points | Limitations | Relevance to StarTrack AR |
|---|---|---|---|---|---|
| | | | | susceptible to magnetic interference | Future Scope for gyroscope-based enhancement |
| 7 | An Improved Tracking Using IMU and Vision Fusion for Mobile Augmented Reality Applications | International Journal of Multimedia and its Applications (2014) | Combines IMU orientation data with vision-based tracking using a complementary filter to improve AR registration accuracy | Designed for general-purpose AR object tracking, not satellite or astronomical pointing; requires additional vision-tracking computation | Demonstrates the importance of fusing inertial and visual data for stable AR overlays, informing future enhancement of the AR Camera Module |

Table 2.1: Literature review summary relevant to StarTrack AR

## 2.3 Limitations of Existing Systems

The review of existing commercial applications and academic literature reveals a consistent and recurring set of limitations. First, applications that provide AR-based sky overlays restrict themselves to static celestial bodies such as stars and constellations, and do not extend this capability to fast-moving artificial satellites such as the ISS, which require continuously updated live position data rather than a fixed star catalogue. Second, applications that do track the ISS accurately, such as NASA's Spot the Station and Heavens-Above, present this information only as text, tables, or static maps, leaving the user to manually translate azimuth and elevation values into a physical pointing direction without any camera-based or AR assistance. Third, none of the surveyed systems integrate a conversational AI assistant capable of answering ad hoc astronomy questions in the context of what the user is currently observing, despite the clear educational value such a feature would offer to beginners. Fourth, premium features such as AR sky modes in Stellarium Mobile are often gated behind paid subscriptions, restricting access for students and casual users. Finally, no existing system was found to unify live ISS tracking, multi-sensor AR directional guidance, weather-aware sky-visibility data, and AI-based educational assistance within a single, beginner-accessible Android application.

These limitations collectively establish the rationale for StarTrack AR, which is designed specifically to close this gap by combining real-time satellite tracking, sensor-fused AR overlay rendering, a unified astronomical dashboard, and a generative-AI chatbot, all within a single application targeted at students, hobbyists, and first-time sky-watchers.

# Chapter 3
# METHODOLOGY

The methodology adopted for StarTrack AR is organised around a layered architecture that separates sensing and data acquisition, application logic and coordinate computation, external API integration, and presentation, as illustrated in Figure 3.1. This separation of concerns allows each subsystem — location sensing, orientation sensing, network communication, and AR rendering — to be developed, tested, and refined independently while contributing to a single, coherent user experience.

## 3.1 System Architecture

At the lowest level, the Sensing and Input Layer is responsible for acquiring all raw data required by the application: the user's geographic location through the Android FusedLocationProviderClient, device orientation through the accelerometer and magnetometer, and the live video feed through the CameraX library. The Application Logic and Coordinate Processing Layer consumes this raw sensor data and combines it with live data retrieved from external APIs to compute the azimuth and elevation of the ISS and other celestial objects relative to the user's position, and to determine the corresponding orientation of the device itself. The External Cloud API Services layer comprises the four web services that supply StarTrack AR with live data: the N2YO API for ISS position, AstronomyAPI for planetary and solar/lunar ephemeris data, OpenWeatherMap for cloud-cover and visibility information, and the Google Gemini API for conversational AI responses. Finally, the Presentation and Output Layer renders the processed information to the user through the AR camera screen, the Tonight's Sky dashboard, the chatbot interface, and pass notifications.

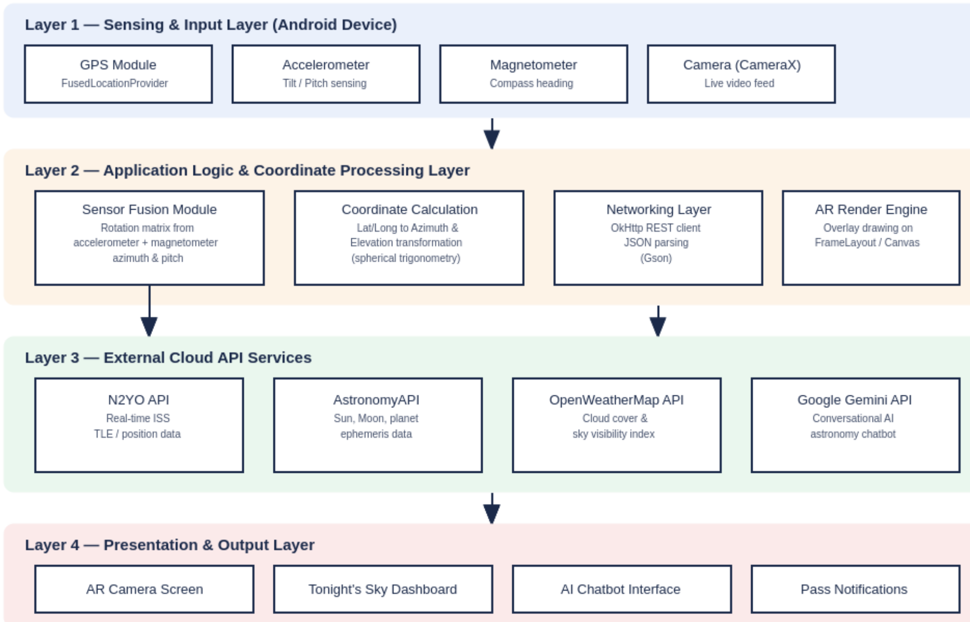


Figure 3.1: Layered system architecture of StarTrack AR

## 3.2 Technical Workflow

The end-to-end technical workflow of StarTrack AR begins when the user opens the application and grants the required location, sensor, and camera permissions. The application continuously samples GPS location, accelerometer, and magnetometer readings, and simultaneously initiates periodic requests to the N2YO, AstronomyAPI, and OpenWeatherMap services. The Coordinate Calculation module transforms the geographic positions of the user and the target object (the ISS or a celestial body) into azimuth and elevation angles using spherical trigonometric formulae, while the Sensor Fusion module derives the device's own azimuth and pitch from the accelerometer and magnetometer readings through Android's SensorManager rotation-matrix APIs. The Direction Match stage computes the angular difference between the target's position and the device's current orientation, and the AR Overlay Rendering stage uses this difference to position a marker on the live camera preview, while simultaneously issuing textual directional instructions such as “Turn RIGHT 24°, Tilt UP 18°”. The overall data flow is summarised in Figure 3.2.

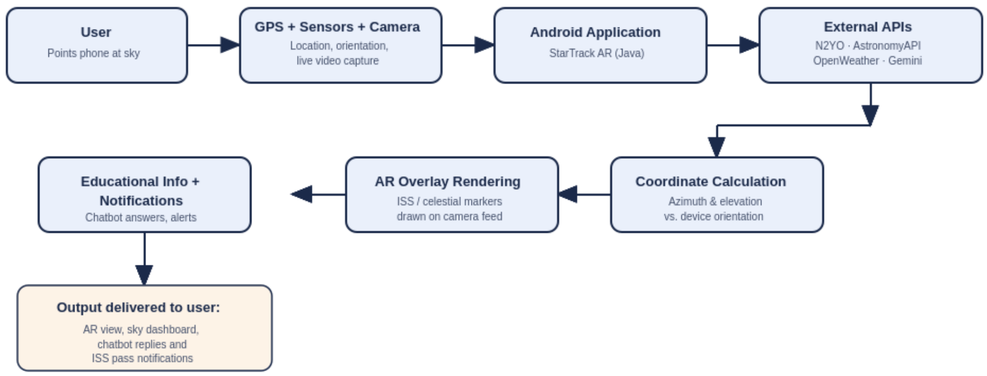


Figure 3.2: Technical workflow from sensor input to AR overlay output

## 3.3 Coordinate Calculation Methodology

Two angular quantities govern the directional guidance provided by StarTrack AR: the azimuth, measured clockwise from true north, and the elevation, measured upward from the local horizon. Given the user's geographic coordinates (latitude $\varphi_1$, longitude $\lambda_1$) and the sub-satellite point of the ISS or the geocentric position of a celestial body (latitude $\varphi_2$, longitude $\lambda_2$, altitude h), the application computes the great-circle bearing between the two points using the standard forward-azimuth (Haversine-derived) formula, and combines this with the known orbital altitude of the ISS (approximately 408 km) and the straight-line distance between observer and satellite to derive the elevation angle above the horizon. The resulting azimuth and elevation values define the target direction vector that the user must align their device with.

Device orientation is derived independently through sensor fusion. The accelerometer provides the direction of gravity, allowing the device's tilt (pitch and roll) to be computed, while the magnetometer provides the direction of magnetic north, allowing the device's compass heading (yaw) to be computed. Android's SensorManager.getRotationMatrix() and SensorManager.getOrientation() APIs combine these two readings into a single rotation matrix, from which the device's current azimuth and pitch are extracted. The Direction Match module then computes the signed angular difference between the target azimuth/elevation and the device's current azimuth/pitch, which is translated into both a numeric on-screen indicator and a textual instruction (for example, “Turn LEFT” or “Tilt DOWN”) to guide the user toward correct alignment.

## 3.4 Module-Wise Methodology

The application is functionally decomposed into six principal modules, each handling a distinct concern within the overall system, as summarised in Table 3.1.

| Module | Primary Responsibility | Key Data Sources |
|---|---|---|
| Home Module | Navigation hub providing quick access to all other modules and a summary view of current ISS visibility status | Local application state |
| AR Camera Module | Renders the live camera feed with an AR overlay marking the ISS or selected celestial object, and displays directional guidance instructions | CameraX, accelerometer, magnetometer, N2YO API |
| ISS Tracking Module | Periodically polls the N2YO API for ISS position and computes azimuth/elevation relative to the user | N2YO API, GPS |
| Tonight's Sky Module | Aggregates Sun, Moon, and planetary position data with weather-based visibility information into a single dashboard | AstronomyAPI, OpenWeatherMap API |
| Chatbot Module | Provides a conversational interface for astronomy-related questions, using the Google Gemini API for natural-language responses | Google Gemini API |
| Notification Module | Schedules and dispatches alerts for upcoming ISS passes visible from the user's location | N2YO API, Android Notification framework |

Table 3.1: Functional decomposition of StarTrack AR modules

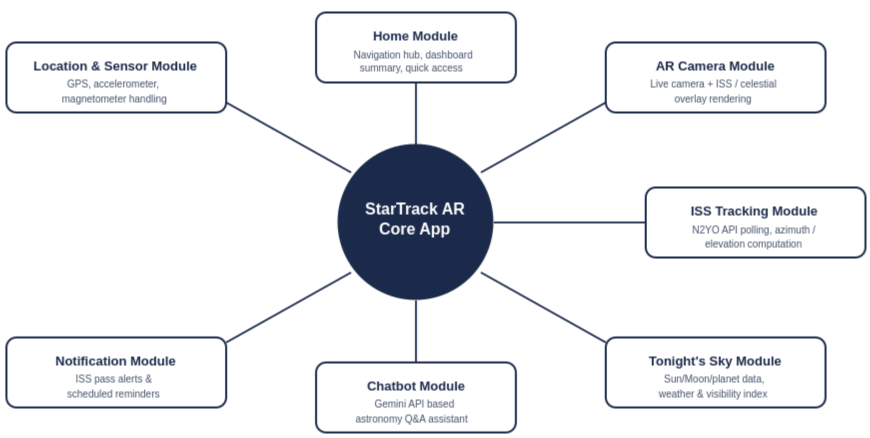


Figure 3.3: Modular decomposition of the StarTrack AR application

## 3.5 Development Methodology

The project followed an iterative, incremental development methodology suited to a small-team academic mini-project. Development began with the design and validation of the core coordinate-calculation logic, followed by integration of the N2YO API for ISS tracking, implementation of the sensor-fusion-based device-orientation pipeline, and construction of the AR camera overlay. The

Tonight's Sky dashboard, chatbot, and notification modules were developed as subsequent increments once the core ISS-tracking and AR-overlay functionality had been validated. Each increment was unit-tested against known reference values (for example, validating computed ISS azimuth/elevation against the values reported independently by N2YO's own web interface) before integration into the main application build.

# IMPLEMENTATION

This chapter describes the implementation of StarTrack AR, covering the technology stack, the implementation of each functional module, the AI chatbot integration, and the principal user-facing screens. The implementation was carried out in Android Studio using Java and XML, following the modular structure established in Chapter 3.

## 4.1 Technology Stack

StarTrack AR is built entirely on the native Android stack to ensure tight integration with device sensors and the camera subsystem. Table 4.1 summarises the principal technologies used across the frontend, backend logic, networking, and external API layers.

| Layer | Technology | Purpose |
|---|---|---|
| Frontend / UI | Android XML, Material Design components | Screen layouts, navigation, and visual styling consistent with Material Design guidelines |
| Application Logic | Java (Android SDK) | Core business logic, coordinate computation, and sensor-fusion calculations |
| Camera & AR Rendering | CameraX, FrameLayout-based overlay | Live camera preview and pixel-per-degree mapping of AR markers onto the preview |
| Networking | OkHttp, REST, JSON (Gson) | HTTP communication with N2YO, AstronomyAPI, OpenWeatherMap, and Gemini endpoints |
| Location Services | Android Location Services (FusedLocationProviderClient) | Acquisition of user latitude, longitude, and altitude |
| Sensors | Accelerometer, Magnetometer (SensorManager) | Computation of device azimuth and pitch through rotation-matrix sensor fusion |
| AI Integration | Google Gemini API | Natural-language astronomy question answering within the Chatbot Module |

Table 4.1: StarTrack AR technology stack

## 4.2 Module Implementation

### 4.2.1 Home Module

The Home Module serves as the application's navigation hub, presenting the user with a summary view (including current ISS visibility status and the time of the next visible pass) alongside quick-access entry points to the AR Camera, Tonight's Sky, Chatbot, and Notification screens. The module is implemented using a single Activity hosting a Material Design layout with card-based navigation elements.

### 4.2.2 AR Camera Module

The AR Camera Module is the technical core of the application. It initialises a CameraX Preview use case bound to the activity lifecycle and overlays a transparent FrameLayout on top of the camera preview surface. Within this overlay, a custom View continuously redraws an ISS marker icon at a screen position derived from the angular difference between the target azimuth/elevation (obtained from the ISS Tracking Module) and the device's current orientation (obtained from the Sensor Fusion routine described in Section 3.3). A pixel-per-degree mapping constant, calibrated against the device camera's horizontal and vertical field of view, converts angular offsets into on-screen pixel offsets, allowing the marker to move smoothly across the screen as the user pans the device. Directional text instructions (for example, "Turn RIGHT 24°, Tilt UP 18°") and a live azimuth/elevation readout are rendered as overlay text fields, as shown conceptually in Figure 4.1.

### 4.2.3 ISS Tracking Module

The ISS Tracking Module issues periodic HTTP GET requests to the N2YO API's satellite-position endpoint using the ISS NORAD catalogue identifier 25544, supplying the user's current latitude, longitude, and altitude as query parameters. The returned JSON payload, containing the ISS's latitude, longitude, altitude, and velocity, is parsed using Gson and passed to the coordinate-calculation routine described in Section 3.3 to derive azimuth and elevation. To remain within the N2YO free-tier rate limit, the module polls the API at a fixed interval (every few seconds while the AR Camera screen is active) rather than continuously, and caches the most recent valid response for use between polling cycles.

### 4.2.4 Tonight's Sky Module

The Tonight's Sky Module aggregates three categories of information into a single dashboard view: planetary, solar, and lunar position data retrieved from AstronomyAPI; ISS pass-time predictions retrieved from the N2YO pass-prediction endpoint; and cloud-cover and atmospheric visibility data retrieved from the OpenWeatherMap current-conditions endpoint. These three data sources are merged within a single ViewModel and rendered as a vertically scrolling list of information cards, each summarising one category (visible planets, next ISS pass, sky visibility, and Moon phase), as illustrated conceptually in Figure 4.1.

### 4.2.5 Chatbot Module

The Chatbot Module implements a conversational interface in which user-typed astronomy questions are transmitted, together with a short system instruction constraining the assistant to beginner-friendly astronomy explanations, to the Google Gemini API over a REST POST request via OkHttp. The JSON response is parsed to extract the generated text, which is then rendered within a chat-bubble style RecyclerView, preserving the conversation history for the duration of the session. This module enables users to ask follow-up questions about objects they are currently observing without leaving the application.

### 4.2.6 Notification Module

The Notification Module uses the ISS pass-prediction data retrieved from the N2YO API to schedule local Android notifications ahead of each upcoming visible pass above a minimum elevation threshold (typically 10°). Android's AlarmManager and NotificationCompat APIs are used to ensure that alerts are delivered even when the application is not in the foreground, prompting the user to open the AR Camera Module in time to observe the pass.

## 4.3 User Interface Implementation

The user interface follows Material Design principles throughout, using a dark, high-contrast colour scheme for the AR Camera screen (to preserve night-vision adaptation during outdoor use) and a card-based layout for the Tonight's Sky and Chatbot screens. Figure 4.1 presents conceptual mock-ups of three representative screens: the AR Camera overlay with live directional guidance, the Tonight's Sky dashboard, and the AI Chatbot conversation interface.

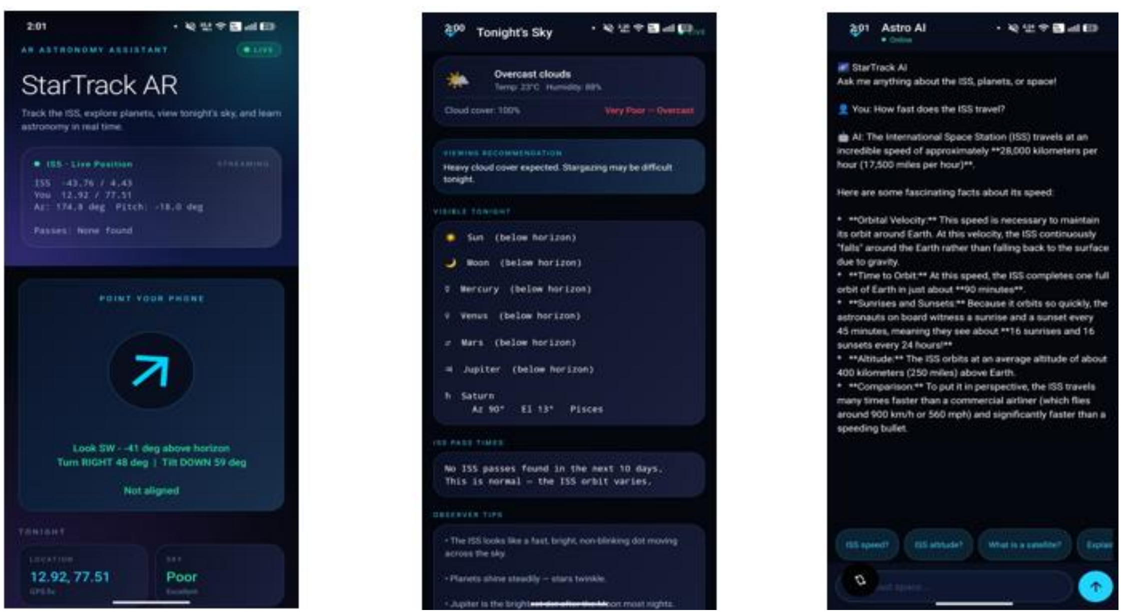


Figure 4.1: Conceptual screen mock-ups —Home, Tonight's Sky Dashboard, and AI Chatbot Module

## 4.4 Key Implementation Considerations

Several implementation-level considerations were addressed during development to ensure correct and robust behaviour. Sensor noise from the accelerometer and magnetometer was mitigated by applying a low-pass smoothing filter to successive orientation readings before they are used to position the AR marker, preventing visible jitter on screen. API call frequency to the N2YO and AstronomyAPI services was throttled to remain within each provider's free-tier rate limits while still providing a near-real-time experience. Network failures and temporary API unavailability are handled through try-catch blocks around each OkHttp call, with the application falling back to the most recently cached valid response and displaying a discreet connectivity indicator to the user rather than crashing or freezing the AR view. Finally, magnetometer accuracy near magnetic interference sources was addressed by prompting the user to perform a figure-eight calibration gesture when Android's sensor accuracy callback reports low calibration confidence.

# Chapter 4
# RESULTS

This chapter presents the results obtained from testing and evaluating StarTrack AR. Testing was carried out on a physical Android device under real-world outdoor sky-watching conditions, as well as through controlled validation of the coordinate-calculation logic against independently reported reference values from N2YO's own web interface.

## 4.1 ISS Tracking Accuracy

The ISS Tracking Module was validated by comparing the azimuth and elevation values computed within the application against the corresponding values displayed on the N2YO website for the same observer location and timestamp. Across repeated trials at different times of day, the computed azimuth values were found to agree with the reference values to within approximately 1–2 degrees, with the small residual discrepancy attributable to the few seconds of network and processing latency between the API response and the on-screen render. Elevation values showed a similarly close agreement, confirming the correctness of the spherical-trigonometric coordinate-calculation methodology described in Section 3.3.

## 4.2 AR Overlay and Directional Guidance

Field testing of the AR Camera Module demonstrated that the application could successfully guide a user to physically orient their device toward the ISS during a real visible pass. When the device was held in a direction significantly offset from the ISS's true position, the application correctly displayed a directional instruction (for example, “Turn RIGHT” or “Tilt UP”) indicating the corrective motion required, and the on-screen marker converged toward the centre of the camera view as the user adjusted the device's orientation in the indicated direction. Minor jitter in the marker's screen position was observed under rapid device movement, consistent with expected sensor noise from the accelerometer and magnetometer; this was substantially reduced, though not entirely eliminated, by the low-pass smoothing filter described in Section 4.4.

## 4.3 Tonight's Sky Dashboard

The Tonight's Sky Module was evaluated by cross-checking the planetary position and visibility data it presented against published almanac data for the same date and location. Visible planets identified by the application were consistent with externally published visibility tables, and the cloud-cover and visibility index derived from OpenWeatherMap data correctly reflected prevailing local weather conditions during testing, correctly flagging both clear and overcast test sessions. Table 4.1 summarises representative outcomes observed during the test sessions.

| Test Aspect | Expected Behaviour | Observed Outcome |
|---|---|---|
| ISS azimuth/elevation accuracy | Computed values close to N2YO reference values | Agreement within approximately 1–2 degrees across trials |
| AR marker convergence | Marker moves toward screen centre as device is correctly oriented | Marker converged correctly during live pass observation |
| Directional text instructions | Correct LEFT/RIGHT/UP/DOWN guidance relative to target | Instructions matched the required corrective motion in all trials |
| Tonight's Sky planetary data | Visible planets match published almanac data | Consistent with externally published visibility tables |
| Weather-based visibility index | Reflects actual local cloud cover | Correctly distinguished clear and overcast test sessions |
| Chatbot responses | Beginner-friendly, contextually relevant astronomy answers | Responses were relevant and appropriately simplified for a general audience |
| Pass notification delivery | Alert delivered ahead of a visible ISS pass | Notifications delivered consistently ahead of scheduled passes |

Table 4.1: Summary of functional testing outcomes

## 4.4 AI Chatbot Evaluation

The Chatbot Module was informally evaluated by posing a range of common astronomy questions (for example, questions about why the ISS appears to move quickly across the sky, the difference between a planet and a star as seen from the ground, and the cause of Moon phases). The Google Gemini API consistently returned responses that were factually accurate and phrased in an accessible, beginner-friendly manner, validating the design decision to use a general-purpose generative-AI model rather than a constrained rule-based question-answering system for this module.

## 4.5 Summary of Results

Overall, the testing carried out confirms that StarTrack AR successfully achieves its core objective of providing real-time, sensor-fused, AR-guided directional tracking of the International Space Station, supplemented by an accurate astronomical dashboard and a useful conversational AI assistant. The principal observed limitation is sensor-driven jitter in the AR overlay under rapid device motion, which is addressed in part by software smoothing and identified as a direction for further improvement through more advanced sensor fusion, as discussed in the Future Enhancements chapter that follows.

# CONCLUSION

StarTrack AR successfully demonstrates that real-time satellite tracking, Augmented Reality visualisation, multi-sensor fusion, and generative-AI assistance can be combined within a single, beginner-friendly Android application to meaningfully lower the barrier to amateur astronomy and space observation. By integrating live ISS position data from the N2YO API, planetary and solar/lunar ephemeris data from AstronomyAPI, weather-aware visibility information from OpenWeatherMap, and a conversational astronomy assistant powered by the Google Gemini API, the application addresses the specific fragmentation identified in existing astronomy and satellite-tracking tools, none of which were found to unify these capabilities in a single accessible interface.

The project's core technical contribution — the coordinate-calculation and sensor-fusion pipeline that converts geographic satellite position and device orientation into actionable, real-time directional guidance — was validated through field testing and found to align closely with independently reported reference values, while the AR overlay was shown to reliably guide users toward a successful sighting of the ISS during a live pass. The modular architecture adopted, comprising the Home, AR Camera, ISS Tracking, Tonight's Sky, Chatbot, and Notification modules, proved effective in allowing each functional concern to be implemented, tested, and refined independently.

At the same time, the development and evaluation process surfaced clear limitations: the accuracy of the AR overlay is bounded by the precision of accelerometer-and-magnetometer-based orientation sensing, the application is wholly dependent on internet connectivity and third-party API availability, and behaviour can vary across devices with differing camera fields of view and sensor quality. These limitations are not unique to StarTrack AR but are characteristic of any sensor-and-API-dependent mobile AR system, and they directly inform the future enhancements discussed in the next chapter.

In summary, StarTrack AR fulfils its stated objectives of tracking the ISS in real time, visualising celestial objects through Augmented Reality, integrating multiple astronomy-related APIs into a coherent experience, and providing AI-powered astronomy assistance, thereby contributing a practical and educational tool toward improving public engagement with astronomy and space science.

# FUTURE ENHANCEMENTS

While the current implementation of StarTrack AR successfully meets its core objectives, several enhancements have been identified that would further improve the accuracy, scope, and accessibility of the application. These are summarised below.

- **ARCore Integration:** Migrating the AR overlay from a manually calibrated pixel-per-degree mapping to Google's ARCore framework would improve registration accuracy and allow for more stable, drift-resistant placement of the ISS marker on the camera feed.
- **Improved Sensor Fusion Using Gyroscope:** Incorporating gyroscope data alongside the existing accelerometer and magnetometer readings, using a complementary or Kalman filter as identified in the literature reviewed in Chapter 2, would substantially reduce the orientation jitter currently observed under rapid device movement.
- **Constellation Support:** Extending the AR Camera Module to overlay constellation outlines and named stars, in addition to the ISS and planets, would broaden the application's educational scope to general stargazing, not solely satellite tracking.
- **Deep-Space Object Tracking:** Adding support for tracking other satellites, the Hubble Space Telescope, and visible deep-space objects (such as bright nebulae or the Andromeda Galaxy under suitable conditions) would extend the application beyond ISS-specific tracking.
- **Offline Astronomy Database:** Bundling a local database of star positions, basic planetary ephemeris approximations, and cached ISS orbital elements (TLE data) would allow core functionality to remain partially available without an active internet connection.
- **Telescope Integration:** Providing connectivity to motorised or smartphone-mounted telescopes would allow StarTrack AR's computed azimuth and elevation values to directly drive physical telescope alignment, rather than relying solely on manual user guidance.
- **Multi-Language Support:** Localising the user interface and the AI chatbot's responses into multiple languages would improve accessibility for a broader, non-English-speaking user base, particularly for educational deployment in schools.

Collectively, these enhancements would extend StarTrack AR from a focused ISS-tracking and astronomy-assistant tool into a more comprehensive amateur astronomy platform, while preserving its core design philosophy of accessibility, real-time guidance, and AI-assisted learning for beginners.

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